WORKSHEET

Chapter 5: Derivatives of vectors

1 Find $\dot{\mathbf{r}}(t)$ and $\ddot{\mathbf{r}}(t)$ for each of the following:

a $\mathbf{r}(t) = 2t^2\mathbf{i} + 5t\mathbf{j}$

b $\mathbf{r}(t) = 7t\mathbf{i} - (6t + 5)\mathbf{j}$

c $\mathbf{r}(t) = 3t^3\mathbf{i} - 9t\mathbf{j}$

d $\mathbf{r}(t) = e^t\mathbf{i} - e^{5t}\mathbf{j}$

e $\mathbf{r}(t) = 2\sin(t)\mathbf{i} + \cos(t)\mathbf{j}$

f $\mathbf{r}(t) = 3\cos(t)\mathbf{i} - 8\sin(t)\mathbf{j}$

2 Use your answers to Question 1 to calculate $\left|\dot{\mathbf{r}}(t)\right|$ and $\left|\ddot{\mathbf{r}}(t)\right|$ for each part.

a

b

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c

d

e

f

3 Find $\dot{\mathbf{r}}(t)$ for each of the following:

a $\mathbf{r}(t) = \frac{5}{3}t^3\mathbf{i} + (3t^2 - 9)\mathbf{j}$

b $\mathbf{r}(t) = (3t + 2)^4\mathbf{i} + 5t^3\mathbf{j}$

c $\mathbf{r}(t) = 9e^{-6t}\mathbf{i} + 4e^{4t-7}\mathbf{j}$

d $\mathbf{r}(t) = 3e^{4t+5}\mathbf{i} - (6e^{\frac{t}{2}+3})\mathbf{j}$

e $\mathbf{r}(t) = \sin(5t)\mathbf{i} - (2t - 5)^2\mathbf{j}$

f $\mathbf{r}(t) = \tan(t)\mathbf{i} + \cos^2(t)\mathbf{j}$

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4 Given $\mathbf{r}(t) = (5t - 3)^6\mathbf{i} - (3t)^3\,\mathbf{j}$, find $\dot{\mathbf{r}}(t)$ and $\ddot{\mathbf{r}}(t)$.

5 If $\mathbf{u}(t) = (2t^2 - 5t)\mathbf{i} + t^3\mathbf{j}$ and $\mathbf{v}(t) = 4t^2\mathbf{i} - t\mathbf{j}$, verify that:

$$\frac{d}{dt}[\mathbf{u}(t)\,.\,\mathbf{v}(t)] = \dot{\mathbf{u}}\,.\,\mathbf{v}(t) + \mathbf{u}(t)\,.\,\dot{\mathbf{v}}$$

6 If $\mathbf{u}(t) = 5t^2\mathbf{i} + (6t - t^2)\mathbf{j}$ and $\mathbf{v}(t) = (2t + 7)^2\mathbf{i} + 2t^4\mathbf{j}$, verify that:

$$\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \dot{\mathbf{u}} \times \mathbf{v}(t) + \mathbf{u}(t) \times \dot{\mathbf{v}}$$

7 If $\mathbf{u}(t) = (5t^3 - 2)\mathbf{i} + (4t - 3t^2)\mathbf{j}$ and $k = 4$, verify that:

$$\frac{d[k\mathbf{u}(t)]}{dt} = k\dot{\mathbf{u}}$$

8 If $\mathbf{u}(t) = 12t\mathbf{i} + (6t^3 - 5t)\mathbf{j}$ and $\mathbf{v}(t) = (7t^5 + 8t^2)\mathbf{i} - 9t^2\,\mathbf{j}$, verify that:

$$\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \dot{\mathbf{u}} + \dot{\mathbf{v}}$$

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9 Find the gradient at the point on the curve determined by the value of t given for each of the following:

a $\mathbf{r}(t) = \tan(t)\mathbf{i} + \sec(t)\mathbf{j}$ at $t = \frac{\pi}{4}$

b $\mathbf{r}(t) = 2t^2\mathbf{i} + 4t\mathbf{j}$ at $t = 3$

c $\mathbf{r}(t) = e^t\mathbf{i} + e^{-2t}\mathbf{j}$ at $t = 2$

d $\mathbf{r}(t) = 2\cos(t)\mathbf{i} - 4\sin(t)\mathbf{j}$ at $t = \frac{\pi}{3}$

e $\mathbf{r}(t) = (t^2 - 2t)\mathbf{i} + (t + 2)\mathbf{j}$ at $t = 4$

f $\mathbf{r}(t) = 3e^{2t}\mathbf{i} + e^{-5t}\mathbf{j}$ at $t = 1$

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Answers

1 a $\dot{\mathbf{r}}(t) = 4t\mathbf{i} + 5\mathbf{j};\ \ddot{\mathbf{r}}(t) = 4\mathbf{i}$

b $\dot{\mathbf{r}}(t) = 7\mathbf{i} - 6\mathbf{j};\ \ddot{\mathbf{r}}(t) = 0$

c $\dot{\mathbf{r}}(t) = 9t^2\mathbf{i} - 9\mathbf{j};\ \ddot{\mathbf{r}}(t) = 18t\mathbf{i}$

d $\dot{\mathbf{r}}(t) = e^t\mathbf{i} - 5e^{5t}\mathbf{j};\ \ddot{\mathbf{r}}(t) = e^t\mathbf{i} - 25e^{5t}\mathbf{j}$

e $\dot{\mathbf{r}}(t) = 2\cos(t)\mathbf{i} - \sin(t)\mathbf{j};\ \ddot{\mathbf{r}}(t) = -2\sin(t)\mathbf{i} - \cos(t)\mathbf{j}$

f $\dot{\mathbf{r}}(t) = -3\sin(t)\mathbf{i} - 8\cos(t)\mathbf{j};\ \ddot{\mathbf{r}}(t) = -3\cos(t)\mathbf{i} + 8\sin(t)\mathbf{j}$

2 a $\left|\dot{\mathbf{r}}(t)\right| = \sqrt{16t^2 + 25};\ \left|\ddot{\mathbf{r}}(t)\right| = 4$

b $\left|\dot{\mathbf{r}}(t)\right| = \sqrt{85};\ \left|\ddot{\mathbf{r}}(t)\right| = 0$

c $\left|\dot{\mathbf{r}}(t)\right| = 9\sqrt{t^4 + 1};\ \left|\ddot{\mathbf{r}}(t)\right| = 18|t|$

d $\left|\dot{\mathbf{r}}(t)\right| = e^t\sqrt{25e^{8t} + 1};\ \left|\ddot{\mathbf{r}}(t)\right| = e^t\sqrt{625e^{8t} + 1}$

e $\left|\dot{\mathbf{r}}(t)\right| = \sqrt{3\cos^2(t) + 1};\ \left|\ddot{\mathbf{r}}(t)\right| = \sqrt{3\sin^2(t) + 1}$

f $\left|\dot{\mathbf{r}}(t)\right| = \sqrt{55\cos^2(t) + 9};\ \left|\ddot{\mathbf{r}}(t)\right| = \sqrt{55\sin^2(t) + 9}$

3 a $\dot{\mathbf{r}}(t) = 5t^2\mathbf{i} + 6t\mathbf{j}$

b $\dot{\mathbf{r}}(t) = 12(3t + 2)^3\mathbf{i} + 15t^2\mathbf{j}$

c $\dot{\mathbf{r}}(t) = -54e^{-6t}\mathbf{i} + 16e^{4t - 7}\mathbf{j}$

d $\dot{\mathbf{r}}(t) = 12e^{4t + 5}\mathbf{i} - \left(3e^{\frac{t}{2} + 3}\right)\mathbf{j}$

e $\dot{\mathbf{r}}(t) = 5\cos(5t)\mathbf{i} - 4(2t - 5)\mathbf{j}$

f $\dot{\mathbf{r}}(t) = \dfrac{1}{\cos^2(t)}\mathbf{i} - [2\sin(t)\cos(t)]\mathbf{j}$

4 $\dot{\mathbf{r}}(t) = 30(5t - 3)^5\mathbf{i} - 81t^2\,\mathbf{j};\ \ddot{\mathbf{r}}(t) = 750(5t - 3)^4\mathbf{i} - 162t\mathbf{j}$

5–8 Demonstrations. Substitute for **u** and **v** and show that LHS equals RHS of the equation.

9 a $\dfrac{\sqrt{2}}{2}$

b $\dfrac{1}{3}$

c $-2e^{-8}$

d $\dfrac{2\sqrt{3}}{3}$

e $\dfrac{1}{3}$

f $\dfrac{-5e^{-7}}{6}$

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